

```
# echo anticipatory > /sys/block/sda/queue/scheduler
```

Add `elevator=anticipatory` in the kernel line of `/boot/grub/grub.conf`, if you want to make it your default.

But how can you tune it as per your needs? Well, the primary tunables are under the `/sys/block/sda/queue/iosched/` directory. Some common values that can be tuned are (the values appear automatically under `/sys/block/sda/queue/iosched/` as soon as you change your elevator):

- **antic_expire** – Here you can set the maximum amount of time (in milliseconds) you anticipate a good read (one with a short seek distance from the most recently completed request) before giving up. For example, I can change the `antic_expire` from the default 6 ms to 60 ms, as follows:

```
# cat /sys/block/sda/queue/iosched/antic_expire
6
# echo 60 > /sys/block/sda/queue/iosched/antic_expire
# cat /sys/block/sda/queue/iosched/antic_expire
60
```

You can make it permanent by adding the echo line in your `/etc/rc.local` file.

- **read_expire** – All the read and write IO requests are processed in batches. So this parameter specifies the time within which the read request must be fulfilled or completed before it expires. The time is specified in milliseconds.

```
# cat /sys/block/sda/queue/iosched/read_expire
125
# echo 250 > /sys/block/sda/queue/iosched/read_expire
# cat /sys/block/sda/queue/iosched/antic_expire
250
```

- **write_expire** – they are equivalent to the above, for writes.

```
# cat /sys/block/sda/queue/iosched/write_expire
250
# echo 300 > /sys/block/sda/queue/iosched/read_expire
# cat /sys/block/sda/queue/iosched/antic_expire
300
```

The deadline scheduler

As the name suggests, here all the I/O requests have to be fulfilled in a specified amount of time—before they expire. So when an I/O request enters the queue, it is assigned some time (in milliseconds), within which it has to be fulfilled before the time expires, regardless of its targeted block device. The same applies for the write requests. It will also specify how the maximum number of read requests will be fulfilled before switching to a write request.

In order to switch to this scheduler, use the following command:

```
# echo deadline > /sys/block/sda/queue/scheduler
```

To make it permanent, add `elevator=deadline` in the kernel line of `/boot/grub/grub.conf`.

And here are its tunables (located under the same `/sys/block/sda/queue/iosched/` directory):

- **read_expire** – Here, when a read request first enters the I/O scheduler, it is assigned a deadline that is the current time plus the `read_expire` value, in milliseconds. And, like before, I can change the default value for `read_expire` from 500 ms to 444 ms

```
# cat /sys/block/sda/queue/iosched/read_expire
500
# echo 444 > /sys/block/sda/queue/iosched/read_expire
# cat /sys/block/sda/queue/iosched/antic_expire
444
```

- **write_expire** – As you might have already guessed, its function is the same as `read_expire`, but it is for writes. The time is again in milliseconds.

```
# cat /sys/block/sda/queue/iosched/write_expire
5000
# echo 4000 > /sys/block/sda/queue/iosched/read_expire
# cat /sys/block/sda/queue/iosched/antic_expire
4000
```

- **front_merges** – Normally, I/O requests are merged at the bottom of the request queue. This Boolean value controls whether an attempt should be made to merge the request to the top of the request queue. A value of 0 indicates that the front merges are disabled.

```
# cat /sys/block/sda/queue/iosched/front_merges
1
```

The Completely Fair Queue scheduler (CFQ)

The CFQ scheduler works in a pure, democratic way. It equally divides all the available bandwidth between all the I/O requests. CFQ uses 64 internal queues to maintain and keep I/O requests. It fills internal queues in a round-robin manner and pulls the I/O requests from these 64 internal queues and places them into a dispatch queue, where they are catered to. Internally, CFQ also changes the order of I/O requests to minimise the head movement.

Most Linux distros come with CFQ as the default I/O scheduler. If not, make it the default as follows:

```
# echo cfq > /sys/block/sda/queue/scheduler
```

Append `elevator=cfq` to the kernel line of `/boot/grub/grub.conf` to make the change permanent.

As for the tunables, they are:

- **quantum** – This is the total number of requests to be moved from internal queues to the dispatch queue in each cycle.

```
# cat /sys/block/sda/queue/iosched/quantum
4
# echo 8 > /sys/block/sda/queue/iosched/read_expire
# cat /sys/block/sda/queue/iosched/antic_expire
8
```

Here I just changed the number of requests to 8 from the default value of 4. Feel free to use `/etc/rc.local` to make the changes permanent.

- **queued** – This is the maximum number of requests allowed per internal queue. You can view and change the current value using the `cat` and `echo` commands, respectively, as above.